

**YOUR WELFARE
REMAINS MY PRIORITY**



**FYMDSA D.O.W
NWODO EMMANUEL
(EMMACO)**

BIO 151: THE EUKARYOTIC PROTISTS: FUNGI AND ALGAE

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What are Eukaryotes?

Eukaryotes are organisms possessing complex cells in which the genetic materials are contained in membrane-bound **nuclei** [singular: nucleus]. The name 'eukaryote' derives from the Greek 'Eu' (True) and 'Karyon' (nut, and by extension, nucleus). Hence the name means 'true nucleus'.

Eukaryotes comprise of multicellular organisms: animals, plants, algae and fungi; as well as unicellular organisms such as yeasts. All eukaryotes have a common origin.

General Characteristics of Eukaryotes

1. Eukaryotic cells are usually ***much larger*** than prokaryotic cells.
2. They contain membrane-bound internal structures called ***organelles***.
3. They have a ***cytoskeleton***, made up of microtubules, microfilaments and intermediate filaments, which help in determining the cell's shape and organization.

The Organelles

a. Nucleus: The most evident organelle in eukaryotes is the nucleus. It is bound by a **nuclear envelope**. It houses the DNA. Eukaryotic DNA is divided into several linear bundles called the **chromosomes**. These (the chromosomes) are separated by a microtubular spindle during nuclear division. The DNA of eukaryotes is usually associated with proteins called **histone proteins**.

b. Mitochondria: This is next to the nucleus the most universal organelle in eukaryotic cells. Mitochondria are surrounded by double membrane which contains invaginations/foldings called **cristae**, where aerobic respiration usually occurs. They contain their own DNA.

c. Other organelles include: **Lysosomes** - which contain enzymes for breakdown of the contents of the **food vacuoles**. **Peroxisomes** – that contain the enzyme catalase for the breakdown of hydrogen peroxide (H_2O_2) to H_2O and O_2 , which otherwise will be toxic to the cell. The **endoplasmic reticulum** (ER), the **ribosomes** (attached to the ER) and **Golgi apparatus** – are involved in the manufacture of proteins. **Plastids** – occur as **chloroplasts** in plants and various groups of algae, and contain chlorophyll which is essential in the production of energy in the process known as **photosynthesis**.

The **cell cytoskeleton** are made up of **microfilaments** (which are polymers of the protein **actin**), and **microtubules** (which are composed of the protein **tubulin**.) They contribute to cell structure and cell motility. In plants and fungi, the cell walls are made up of **cellulose** and **chitin**.

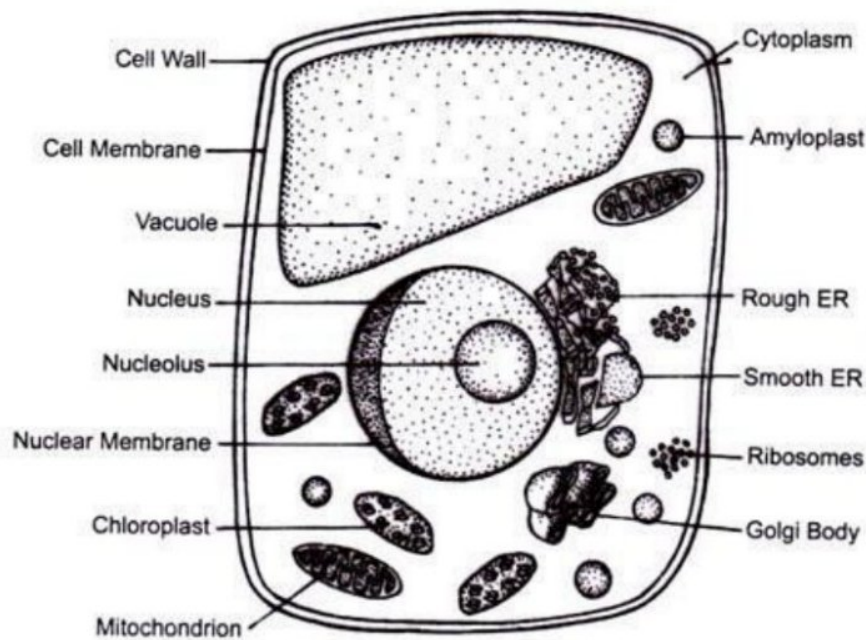


Fig. 2.1 Structure of Plant Cell

Fig 1: Structure of a Eukaryotic Cell (Plant Cell)

Eukaryotes: Overview of Classification

Eukaryotes were commonly classified into four kingdoms: protists (algae and protozoans), fungi, plants and animals. Together with all prokaryotes (lumped together in Kingdom Monera), they formed the so-called five kingdom classification of living organisms.

However, the application of comparative molecular sequencing using **16S ribosomal RNA** has resulted in the re-classification of living organisms into three domains – the **Bacteria**, the **Archaea**, the **Eukarya/Eukaryotes** (i.e., Fungi, Algae, Protozoa, Plants and Animals). The study discovered that eukaryotes are more related to the Archaea than to the bacteria.

FUNGI

Fungi are a large and diverse group of primarily terrestrial spore-forming organisms of about 80,000 named species, and probably a million more which are still unnamed. They range from the **unicellular yeasts** to the large **multicellular toadstools** and **puffballs**. They occupy a wide range of habitats, both aquatic and terrestrial, and have been exploited by man for both economic and medicinal purposes.

Fungi include:

- i. The **moulds** which grow on damp organic matter, e.g., bread, leather, decaying vegetation and dead fish
- ii. The unicellular **yeasts** found on the sugary surface of fruits

iii. Some **parasites of plants** that are responsible for some economically important diseases of crops such as mildew, rusts and smuts. (A few fungi also cause diseases in humans and animals.)

The study of fungi is called **mycology**.

Major Characteristics of Fungi

- a. They may be **unicellular** (yeasts) or **multicellular**.
- b. They are **eukaryotes**.
- c. They consist of long threadlike filaments called **hyphae** [singular: hypha]. Groups of hyphae congregate to form a tangled mass called **mycelium** [plural: mycelia].
- d. The cell walls of fungi are made primarily of a polysaccharide called **chitin**, a polymer of the glucose derivative N-acetylglucosamine. It is also sometimes made of cellulose.
- e. The protoplasm of the hyphae is either continuous or interrupted at intervals by cross-walls **septa** [singular: septum.]
- f. Hyphae separated by these cross walls (septa) are said to be **septate**, while hyphae with no such cross walls are said to be **coenocytic** or **non-septate/aseptate**.
- g. Fungi are **saprophytic**, i.e., they obtain their nutrients from dead decaying organic matter by secreting an exoenzyme and then absorbing the solubilised nutrient.
- h. Some are **dimorphic**, i.e., they can exist as yeasts or as moulds under different environmental conditions.
- i. Reproduction in fungi can be **sexual** or **asexual** by means of **spores** or **conidia**. [Spores are non-motile reproductive cells that rely on being carried by animals or the wind for their dispersal.]
- j. Fungi are primarily **terrestrial** organisms.
- k. They are mostly aerobic, although some are facultatively aerobic.

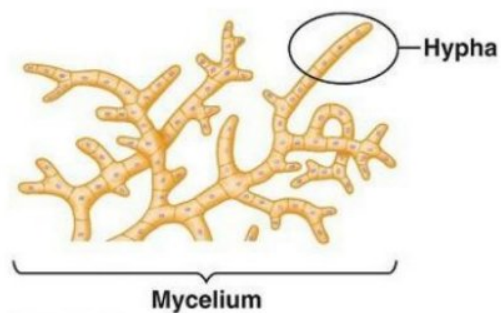


Fig 2 : Hypha and Mycelium

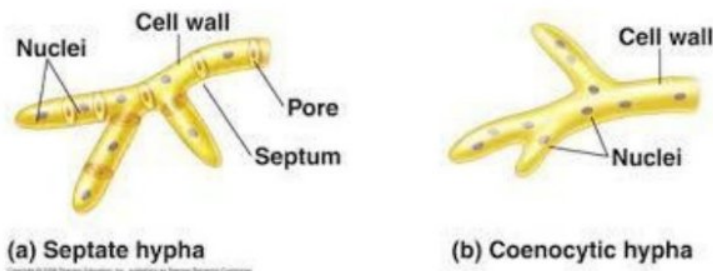


Fig 3: Septate Hypha and Aseptate/Non-Septate or Coenocytic Hypha

Classification of Fungi

Fungi are classified into **five** groups based on the type of sexual spores they produce: Ascomycetes, Zygomycetes, Basidiomycetes, Oomycetes and Deuteromycetes.

1. Ascomycetes:

Common name: Sac fungi

Type of sexual spore present: Ascospore. These are a type of spores formed within an enclosed sac called **ascus**.

Nature of the hyphae: Septate.

Typical members: *Saccharomyces* (yeasts) and *Neurospora*.

About half of the Ascomycetes exist in symbiotic associations with algae called **lichens**.

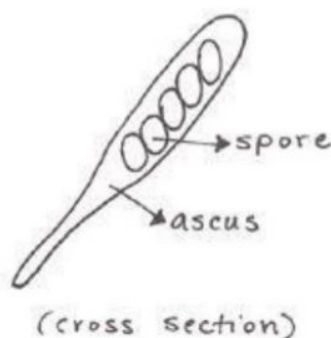


Fig 4: Ascospores in an Ascus

2. Zygomycetes:

Nature of the hyphae: Aseptate (Coenocytic)

Type of sexual spore present: Zygosporangium. Zygosporangia are macroscopically visible and result from the fusion of hyphae and the exchange of genetic material. These zygosporangia are resistant to unfavourable environmental conditions.

Typical members: *Mucor*, *Rhizopus* (black bread mould)

Some of the zygomycetes are of great importance because they form mutualistic associations with plant roots known as **mycorrhiza**.

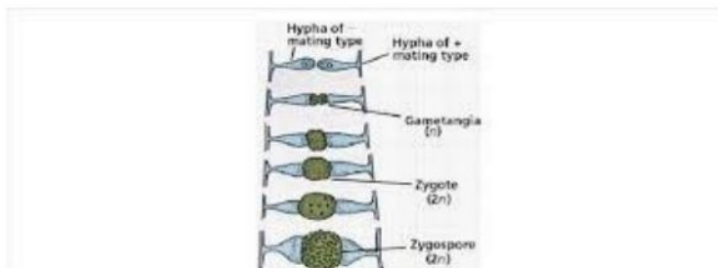


Fig 5: The Formation of a Zygosporangium

3. Basidiomycetes:

Common Name: Club fungi

Nature of the hyphae: Septate

Type of Sexual Spore Present: Basidiospores. These are a type of spores produced on the ends of a club-shaped structure called **basidium** [plural: basidia].

Typical members: Include members of the genera *Agaricus* and *Amanita*. These are the true mushrooms and toadstools, as well as other familiar fungi such as puffballs and bracket fungi. They also include plant pathogens like rusts and smuts.



Fig 6: Different types of basidia and basidiospores

4. Oomycetes:

Common Name: Water Moulds

Nature of Hyphae: Aseptate (Coenocytic).

Type of Sexual Spore Present: Oospore. An oospore is a thick-walled sexual spore that develops from a fertilized egg in some fungi and algae.

Typical members: *Albuginales* (white blister rusts)

The cell walls of Oomycetes are composed of cellulose rather than chitin as in most fungi. Some are water-dwelling, but most are, like other fungi, terrestrial.

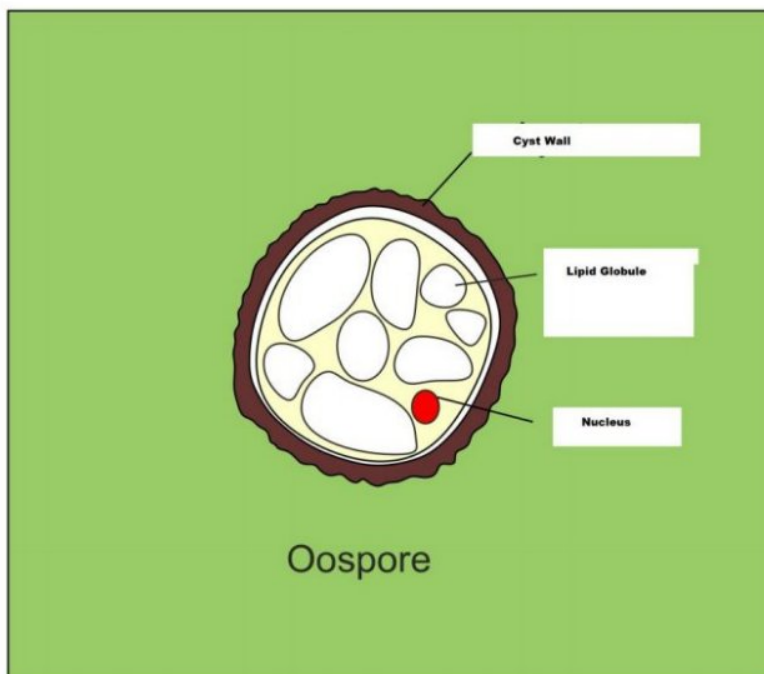


Fig 7: An Oospore

5. Deuteromycetes:

Common Name: Fungi Imperfecti

Nature of Hyphae: Septate.

Type of Sexual Spore Present: None. Only their asexual form of reproduction is known.

Typical Members: *Penicillium*, *Aspergillus*, *Candida*.

ALGAE

Algae are a large heterogeneous group of photosynthetic eukaryotes (some unicellular and some multicellular) that are largely aquatic in nature. A few are terrestrial however. The bodies of algae are poorly developed, lacking a well-developed vascular system (roots, stems, leaves), and complex reproductive structures. They possess chlorophyll and other pigments for carrying out photosynthetic activities. The body of an alga is referred to as a **thallus** [plural: thalli]

A few other general characteristics of algae are:

1. They can be unicellular, colonial, filamentous, tubular or membranous.
2. They reproduce sexually and asexually.

Classification of Algae:

Algae are classified chiefly based on the type of photosynthetic pigments they possess, their types of storage carbohydrates, types and arrangement of flagella, etc. Based on these features algae are classified into the following divisions or **phyla** [singular: **phylum**.]

1. Phylum Chlorophyta:

Common name is **green algae**.

They are either unicellular or multicellular.

The dominant photosynthetic pigment is **chlorophyll** (responsible for the green colour of these algae.) Chlorophylls **a** and **b** are present.

They store carbohydrates as **starch**. The starch is stored in structures called **pyrenoids**.

Their cell walls are composed of cellulose.

They are largely freshwater organisms.

Typical Members: *Chlamydomonas*, *Spirogyra*

2. Phylum Euglenophyta:

Common name is ***Euglenoids***.

They are unicellular organisms.

Their photosynthetic pigment is chlorophyll, and Chlorophylls **a** and **b** are present.

They store carbohydrates as ***paramylon***.

Cellulose cell walls are absent in Euglenoids. Instead they have a plasma membrane, made of flexible ***pellicle***.

Typical Members: *Euglena*

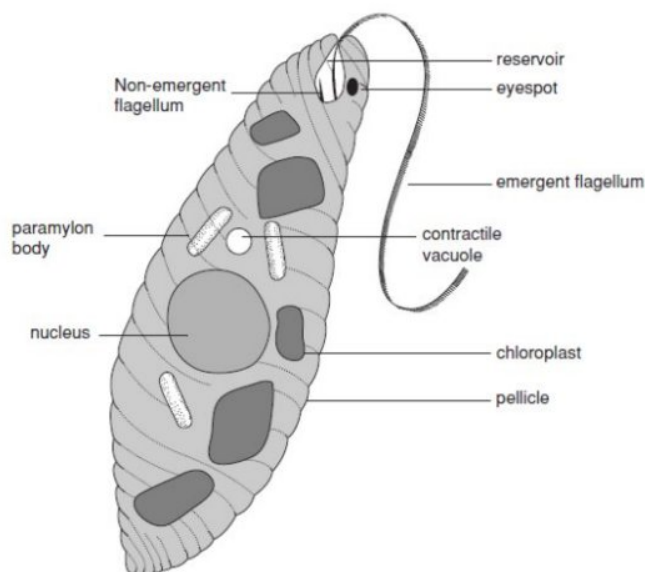


Fig 8: Euglena.

3. Phylum Chrysophyta:

Common names of members of this phylum are Golden-brown algae and diatoms.

They are unicellular organisms.

Their photosynthetic pigment is chlorophyll, and Chlorophylls **a** and **c** are present.

They store carbohydrates as **lipids**.

Their cell walls are composed of cellulose, silica, etc.

Typical Members: *Ochromonas*.

4. Phylum Dinoflagellata/Pyrrophyta

Their common name is dinoflagellates.

They are unicellular organisms.

Their photosynthetic pigments are Chlorophylls a and c, and **Xanthophylls** (yellow pigments), which gives most dinoflagellates a red/golden appearance.

They store carbohydrates as starch.

Their cell walls (when present) are composed of **cellulose**; and the cells are covered with armoured plates known as **thecae** [singular: theca.]

They are largely marine organisms.

Typical Members: *Gonyaulax*, *Alexandrium*, *Pfiesteria*.

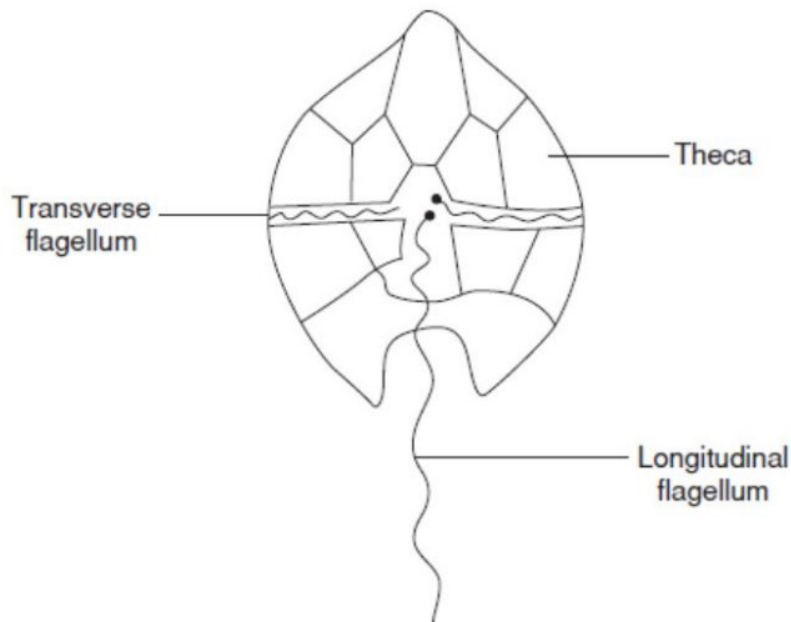


Fig 9: A dinoflagellate

5. Phylum Phaeophyta:

The common name is brown algae.

They are multicellular organisms (large and complex seaweeds).

Their photosynthetic pigments are Chlorophylls **a** and **c** and Xanthophylls (in this case, a type of xanthophylls called **fucoxanthin**, which is a **brown** pigment).

They store carbohydrates as **Laminarin**.

Their cell walls comprises of cellulose.

They are mostly marine organisms.

Typical members: *Laminaria*, *Fucus*.

6. Phylum Rhodophyta:

Their common name is **red algae**.

They are mostly multicellular organisms.

They contain the photosynthetic pigments Chlorophylls **a** and **d**, as well as the pigments phycocyanin and phycoerythrin which give the red algae their typical red colour.

They store carbohydrates as starch, and they have cellulose in their cell walls.

Typical Members: *Palmaria*, *Lemanea*.

Table 1: A Summary of the Classification and Characteristics of Algal Groups

	Common name	Morphology	Pigments	Storage compound	Cell wall
Euglenophyta	Euglenids	Unicellular	Chlorophyll <i>a</i> & <i>b</i>	Paramylon	None
Pyrrophyta	Dinoflagellates	Unicellular	Chlorophyll <i>a</i> & <i>c</i> , xanthophylls	Starch	Cellulose/ none
Chrysophyta	Golden-brown algae, diatoms	Unicellular	Chlorophyll <i>a</i> & <i>c</i>	Lipids	Cellulose, silica, CaCO ₃ , etc.
Chlorophyta	Green algae	Unicellular to multicellular	Chlorophyll <i>a</i> & <i>b</i>	Starch	Cellulose
Phaeophyta	Brown algae	Multicellular	Chlorophyll <i>a</i> & <i>c</i> , xanthophylls	Laminarin	Cellulose
Rhodophyta	Red algae	Mostly multicellular	Chlorophyll <i>a</i> & <i>d</i> , phycocyanin, phycoerythrin	Starch	Cellulose

PROTOZOA

Protozoans are a very diverse group of unicellular eukaryotic organisms. Unlike alga, they are non-photosynthetic and generally colourless, hence they are heterotrophic. Protozoans lack cellulose cell walls and are usually found in freshwater and marine habitats. A large number of them are parasitic in other animals, as well as in humans, being responsible for a range of diseases. Protozoans typically feed by the process known as **phagocytosis**, whereby the food particle is surrounded with a portion of their cell membrane engulfs the particle and brings them into the cell for digestion in the **phagocytic vacuoles**. Other protozoans (e.g., Paramecium) have **gullets** for swallowing smaller bacterial or eukaryotic cells.

The vast majority of protozoans are motile. Hence, protozoans are chiefly classified based on the structural features used to achieve locomotion (cilia, flagella, etc), into Sarcodina, Mastigophora, Ciliophora and Apicomplexa.

1. Sarcodina (The Amoebas):

They are characterised by the possession of **pseudopodia** (= false feet, singular: **pseudopodium**), which are temporary projections from the cell, which the amoebas use for locomotion. The pseudopodia are also used for capturing and engulfing food.

Amoebas reproduce by simple binary fission. Most amoebas are free-living in aquatic environments. A wide variety of amoebas is parasites of humans and other vertebrates, and usually inhabits the oral cavity and the intestinal tract. A good example is *Entamoeba histolytica* which causes amoebic dysentery in humans.

Shelled amoebas (Foraminifera and radiolaria) – Some amoebas have an external shell covering the cell. Examples are the **foraminifera** and the **radiolaria**. The shells are called **tests** and are made of calcium carbonate.

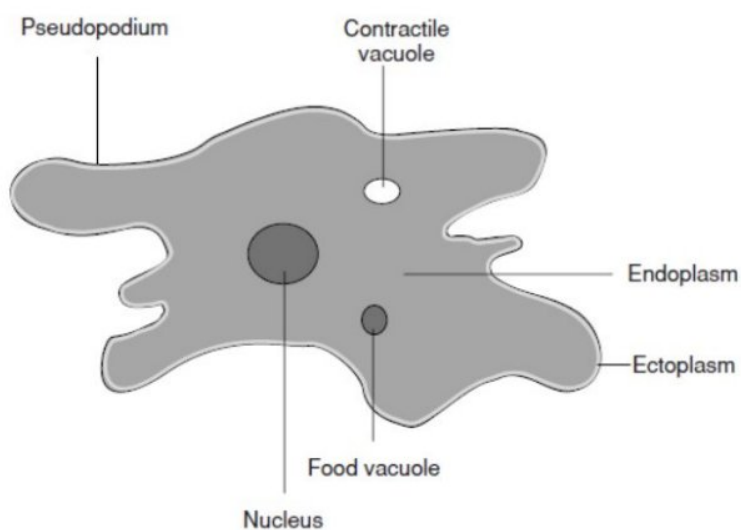


Fig 10: Structure of an Amoeba

2. Mastigophora (The Flagellates or Zooflagellates):

They are characterised by the possession of the long **flagellum** [plural: flagella] by which they propel themselves around. Although many members of this phylum are free-living, some are parasitic in and pathogenic for animals and humans. Examples are the trypanosomes. A member of this genus, *Trypanosoma gambiense* is responsible for African sleeping sickness. It lives in the salivary glands of the intermediate host, the tse tse fly (*Glossina spp*) and is passed to the human host when a bite punctures the skin. It reaches the central nervous system via the blood or lymphatic systems and causes inflammation of the brain and spinal cord resulting in the characteristic lethargy, coma and eventual death of the patient.

Other disease-causing examples of the Mastigophora are – *Giardia lamblia*, causative agent of intestinal disease **giardiasis**; and *Leishmania spp* – causative agent of leishmaniasis.

Reproduction in the mastigophora is generally by binary fission.

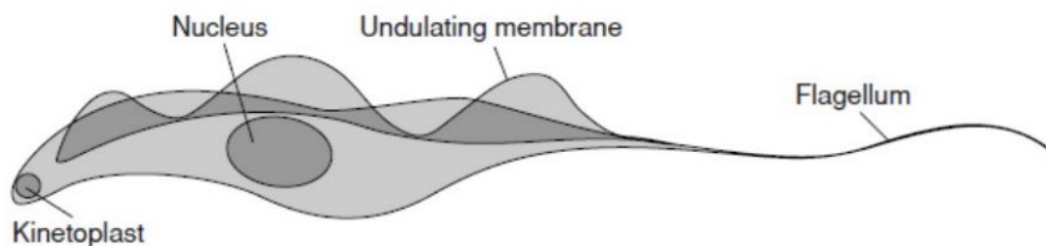


Fig 11: Trypanosoma, an example of Flagellates

3. Ciliophora (Ciliates):

Ciliates are characterised by the possession of **cilia** in some stages of their life cycles. The cilia beat in coordinated fashion to propel the organism or assist in the ingestion of food particles. Ciliates are also unique among protozoans in having two distinct kinds of nuclei:

- **Macronucleus:** Involved only in the production of messenger ribonucleic acid (RNA) for various aspects of cell's essential metabolic process.
- **Micronucleus:** Involved solely in sexual reproduction by conjugation.

Most ciliates possess a mouth structure known as the **cytostome** connected to an underlying gullet through which food is ingested. Undigested food particles are ejected through the anal pore or the **cytoproct**. Many ciliates also have **trichochysts**, which are long filaments that enable the protozoa attach to surfaces and aids it in defence.

The best-known and most widely distributed of the ciliates are members of the genus ***Paramecium***. The only ciliate known to cause disease in humans is ***Balantidium coli***, which causes a form of dysentery.

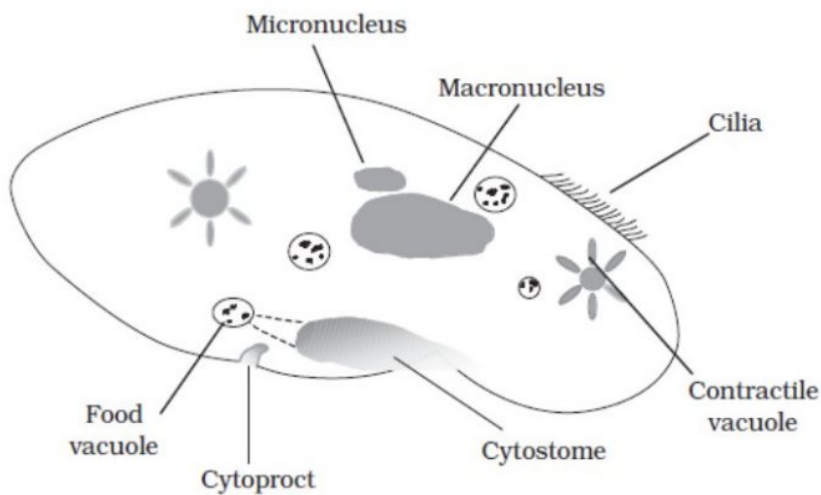


Fig 12: *Paramecium*

4. Apicomplexa (The Sporozoans)

The sporozoans are all obligate parasites. They are generally non-motile. Sporozoans are characterised by a spore-like stage called a sporozoite, which is involved in the transmission of the parasite to a new host.

The best known sporozoans are the ***Plasmodium***, the causative agent of malaria, transmitted to humans by the *Anopheles* mosquito. Another known is ***Toxoplasma***, which is the causative agent of **toxoplasmosis**.

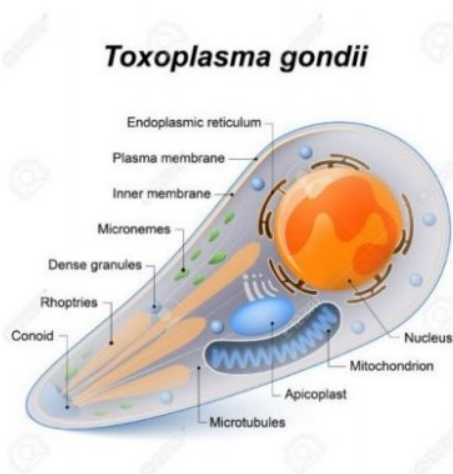


Fig 13: *Toxoplasma gondii*, an example of a Sporozoan